Histology of Blood Vessels

Main functions of Vascular system

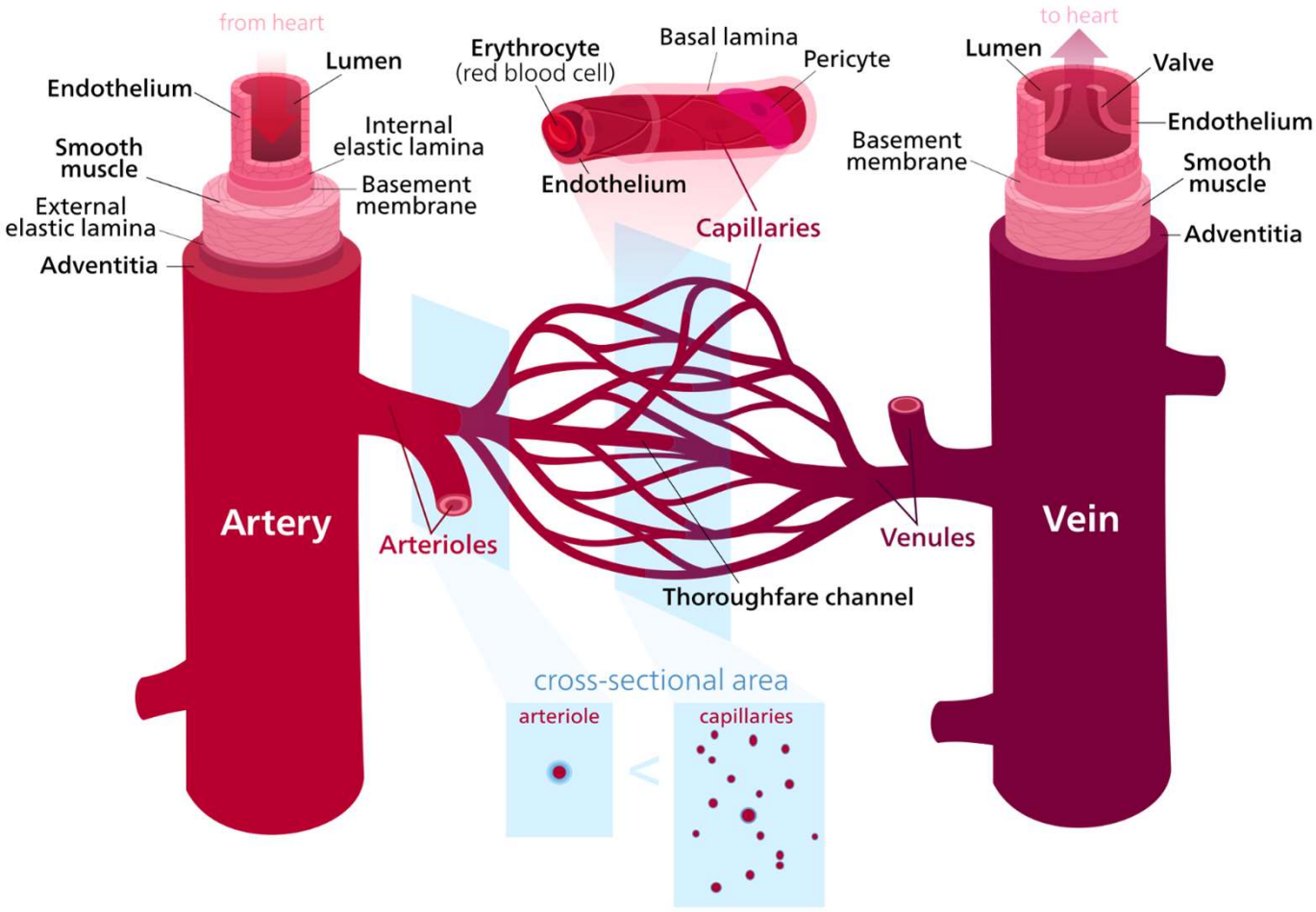
- Transport of oxygen, carbon dioxide, nutrients, hormones, metabolic products, cells of immune defense system and many other essential products

- Gaseous exchange

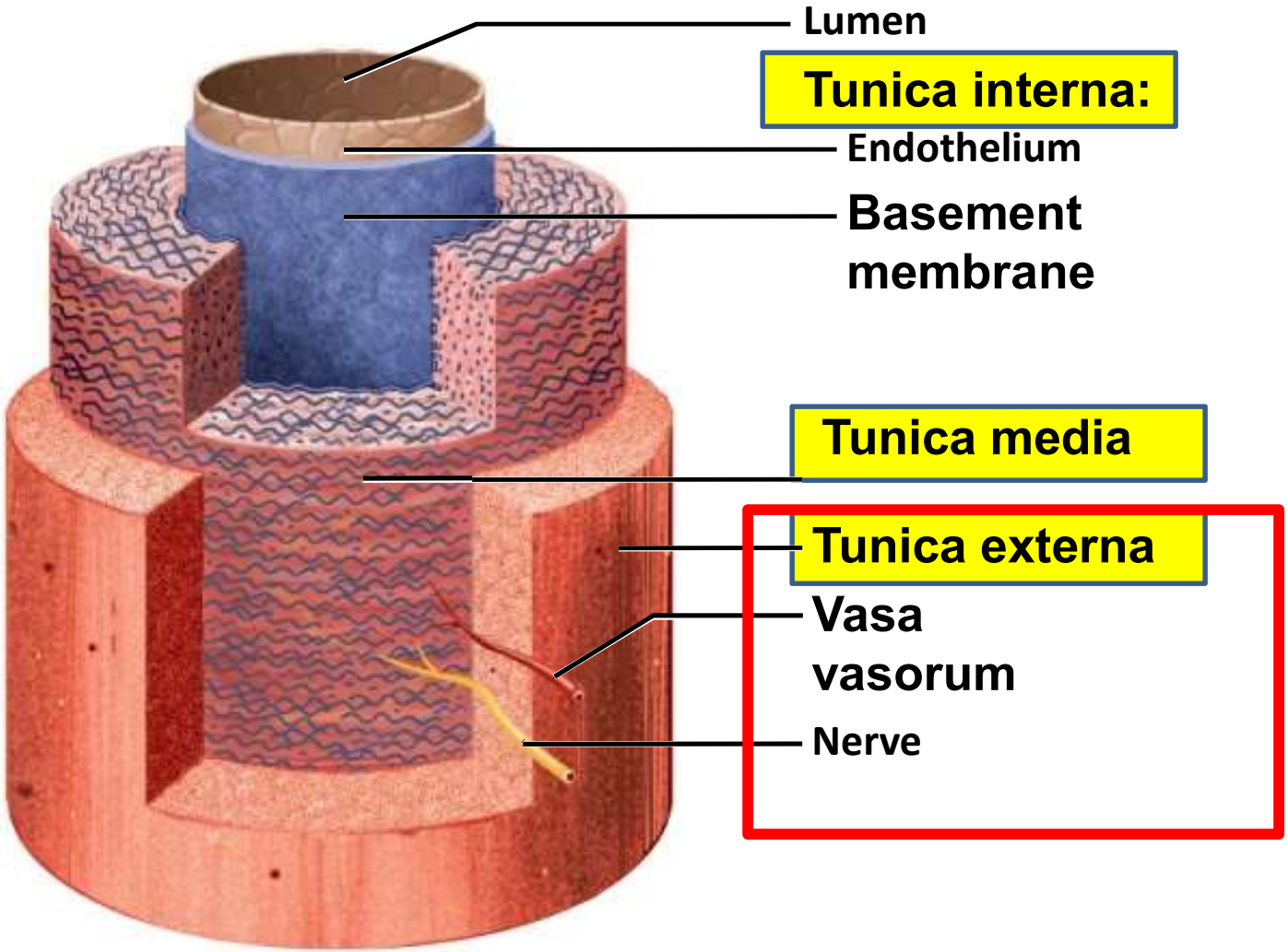
- Temperature control

Types of Blood Vessels

- **Arteries:** ***Large or Elastic Artery***
  Medium sized or Muscular Artery
- **Arterioles**
- **Capillaries**: ***Continuous capillary***
  Fenestrated capillary
  Sinusoidal capillary
- **Venules**
- **Veins:** ***Medium sized Vein***
  Large Vein


from heart
Lumen
Endothelium
Internal elastic lamina
Smooth muscle
Basement membrane
External elastic lamina
Adventitia
Erythrocyte (red blood cell)
Basal lamina
Pericyte
Endothelium
Capillaries
to heart
Lumen
Valve
Endothelium
Basement membrane
Smooth muscle
Adventitia
Artery
Arterioles
Venules
Vein
Thoroughfare channel
cross-sectional area
arteriole
<
capillaries

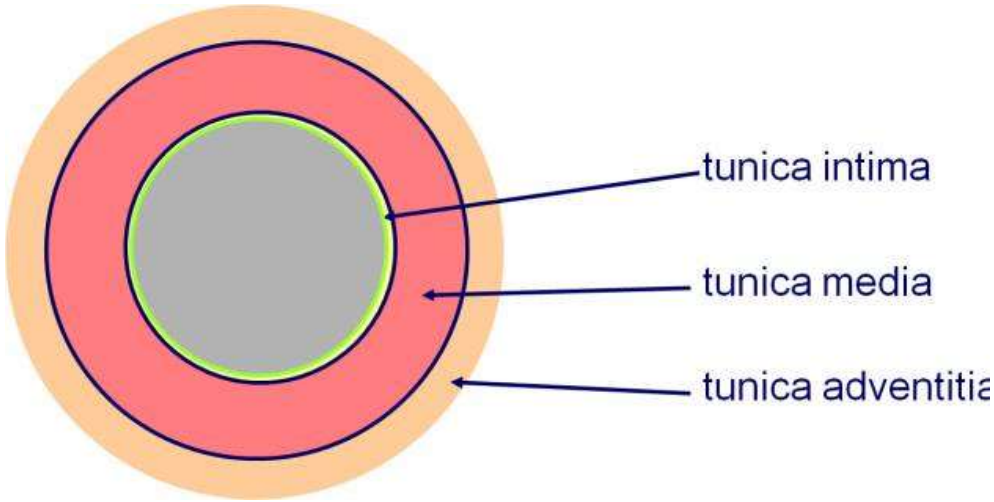
Conducting (large) artery



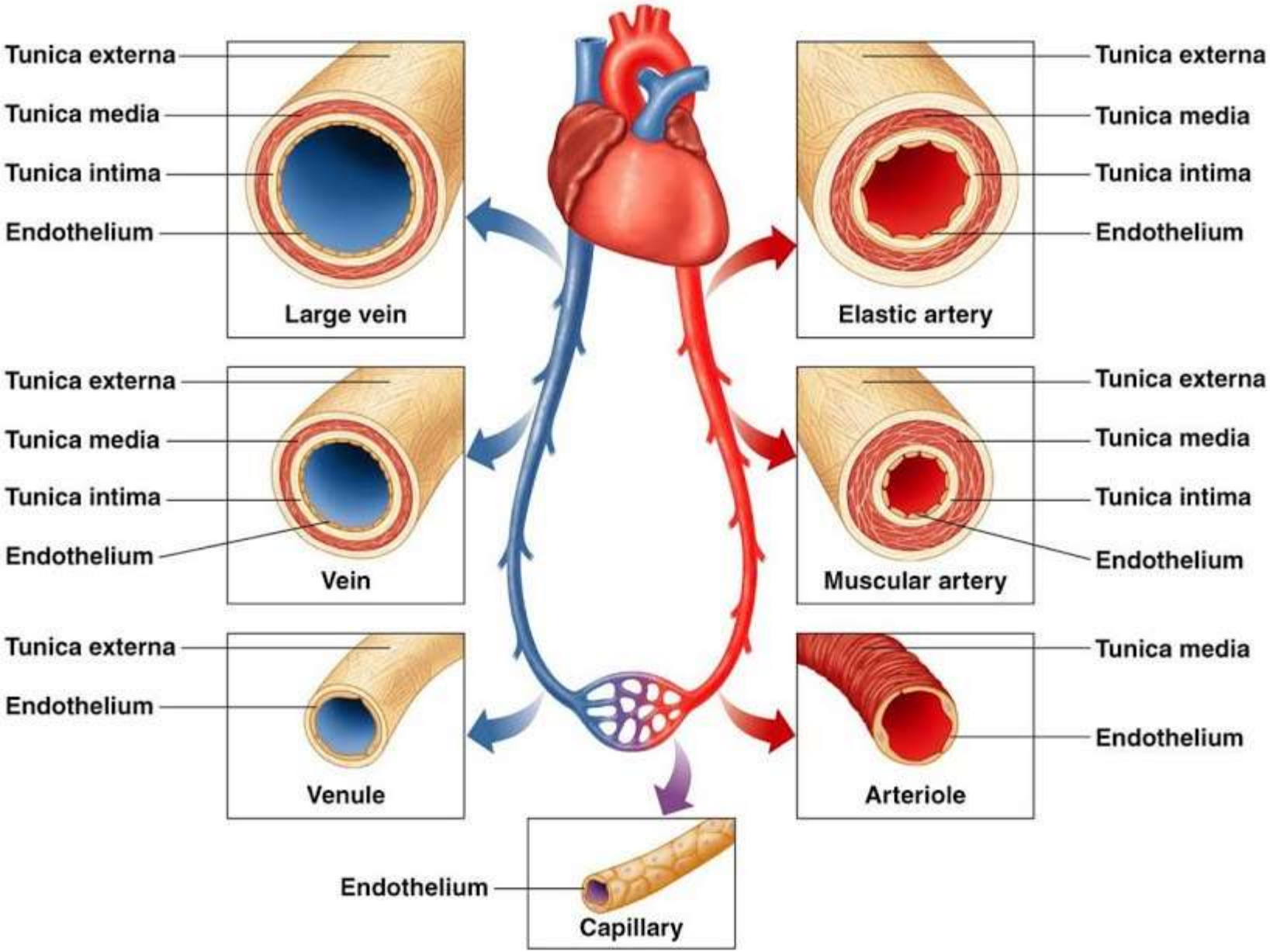
Structure of Blood Vessels

All blood vessels except capillaries have 3 basic Tunics or Coats concentrically arrranged

- **Tunica Intima**
- **Tunica Media**:
- **Tunica Adventitia** :



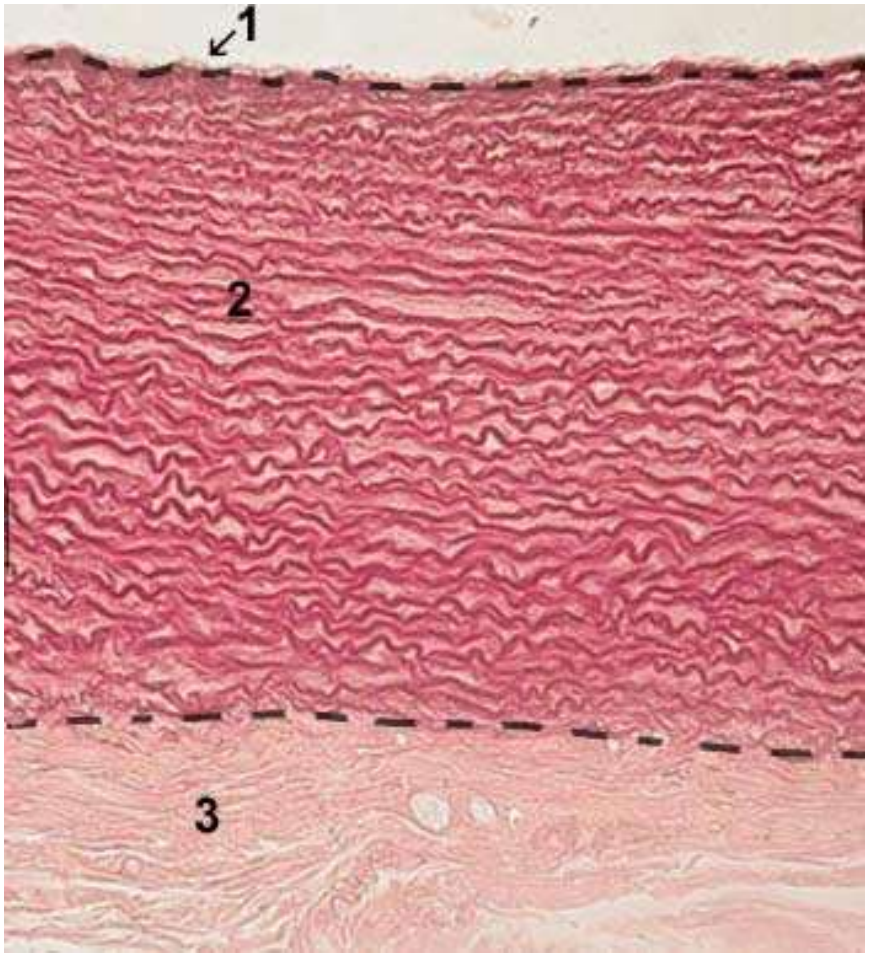
Arteries and Veins



Elastic Artery(stained with orcein)

Tunica Intima:

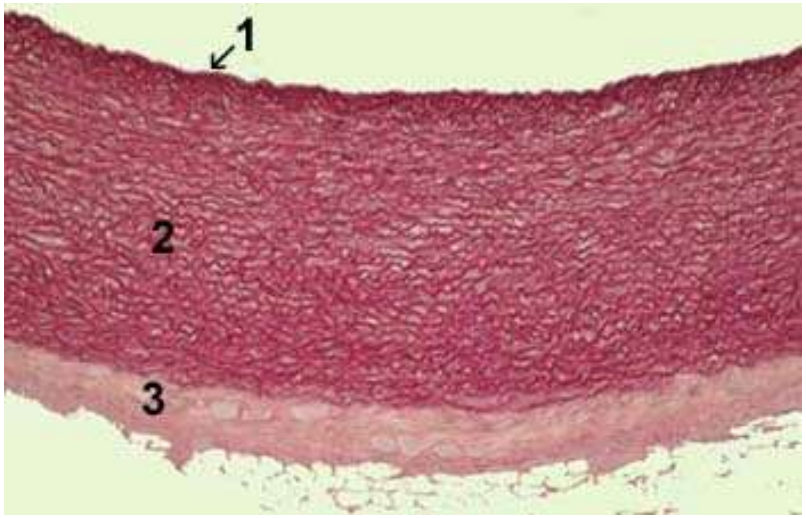
- ***Endothelium***
- ***Basal lamina*** **:A thin layer of glycoprotein**
- ***Subendothelial C T*** **which contains fibrocytes, macrophages and smooth muscle**
- **The collagen and elastic fibres are longitudinally arranged.**
- ***Internal Elastic Lamina*** **forms the boundry between T Intima and T media. Consists of elastic fibres is poorly defined and is fenestrated**



Elastic Artery (stained with orcein)

Tunica Media:

- Elastic tissue in the form of 40-70 fenestrated elastic membranes
- Smooth muscles and collagen fibres in matrix
- Matrix rich in chondroitin sulfate.
- A layer of elastic fibers external elastic lamina forms the boundary between T media and T adventitia.

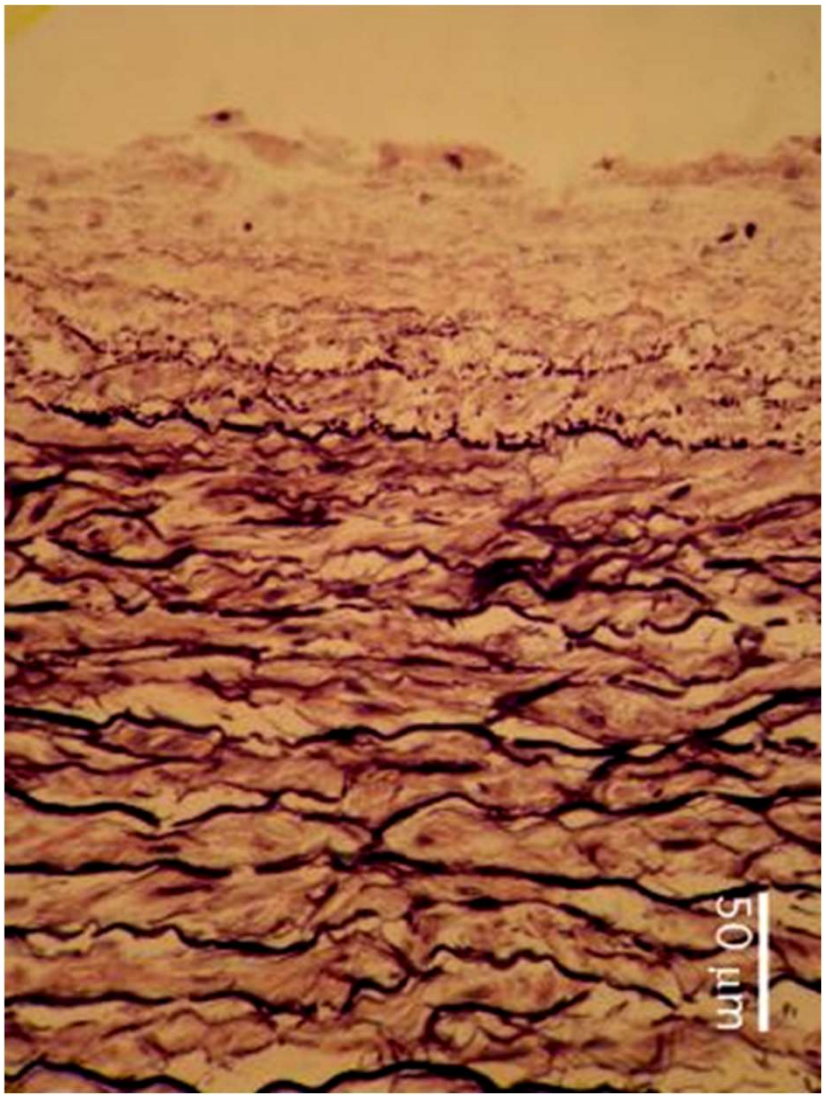


Tunica Adventitia:

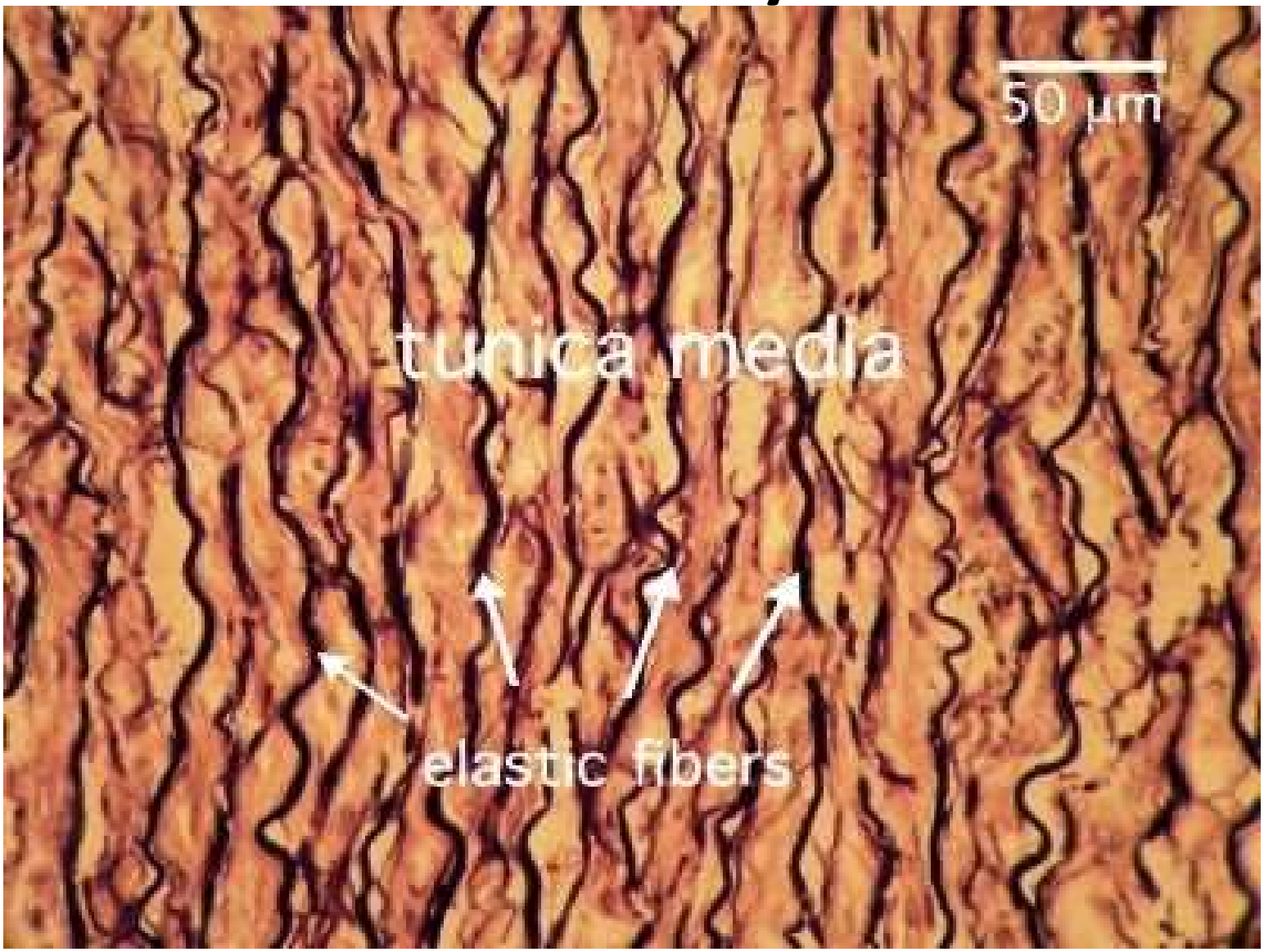
Fibroelastic C T having Vasa Vasorum and unmyelinated sympathetic fibres.

Elastic Artery

- Presence of **elastic fibres** in the wall ***allows it to expand during contraction (systole) and recoil during relaxation (diastole).***
- This maintains **necessary Blood Pressure and permits the blood to flow more evenly** through other arterial channels



Prominent Elastic fibres in T Media of Elastic Artery



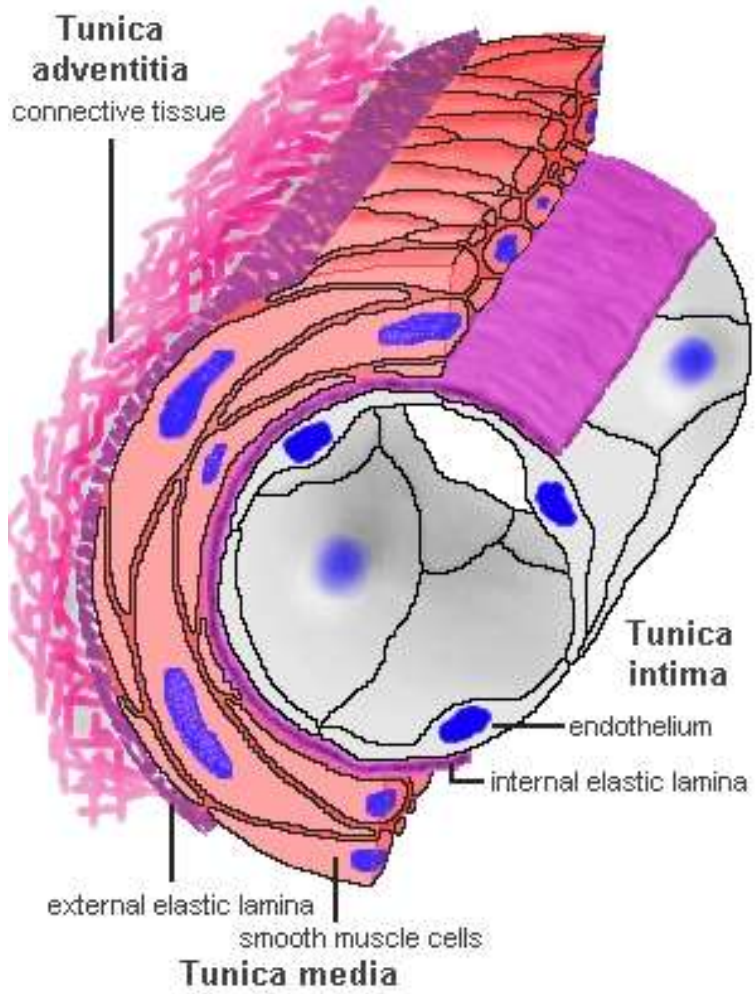
Muscular Artery

- **T Intima:**

  Endothelium

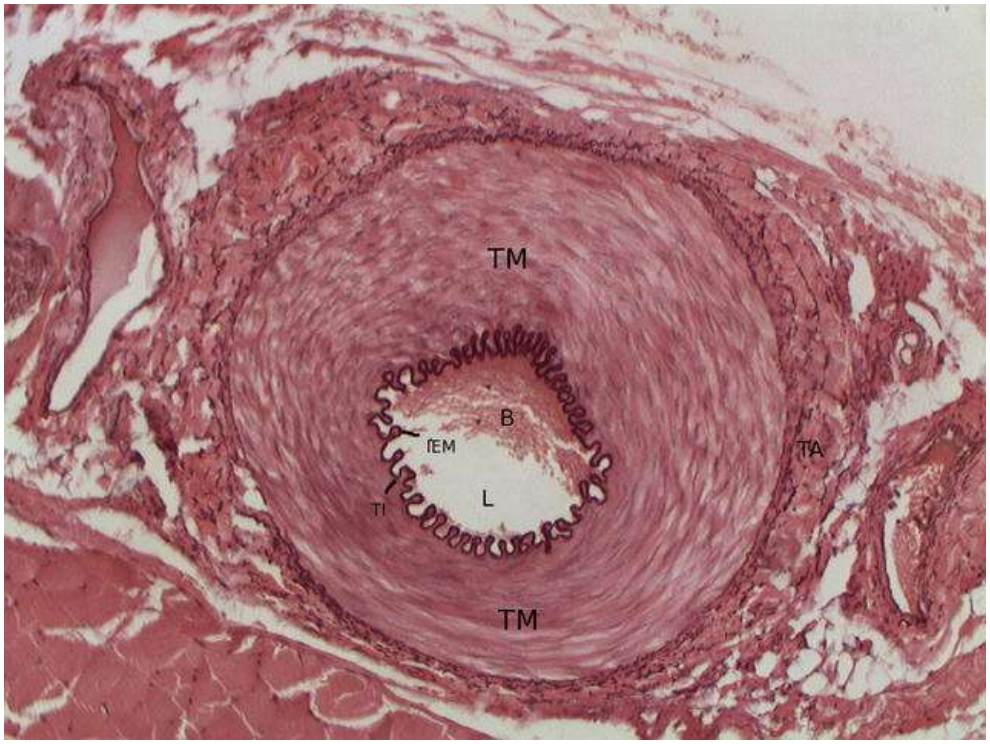
  Internal Elastic lamina (No subendothelial Tissue)

  Internal Elastic Lamina ***is a bright refractile membrane thrown into wavy folds due to contraction of the elastic membrane during the fixation process.***



Muscular Artery

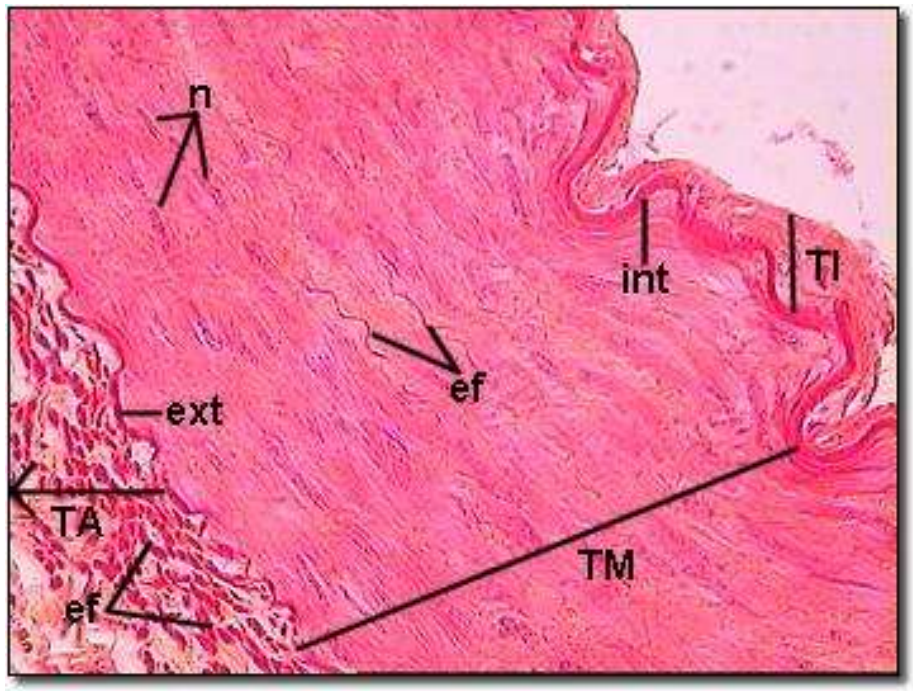
- **T Media**: ***40 layers of smooth muscle cells arranged circularly, hence the name muscular artery.***
- Contains **few collagen and elastic fibres** intermixed with smooth muscle cells



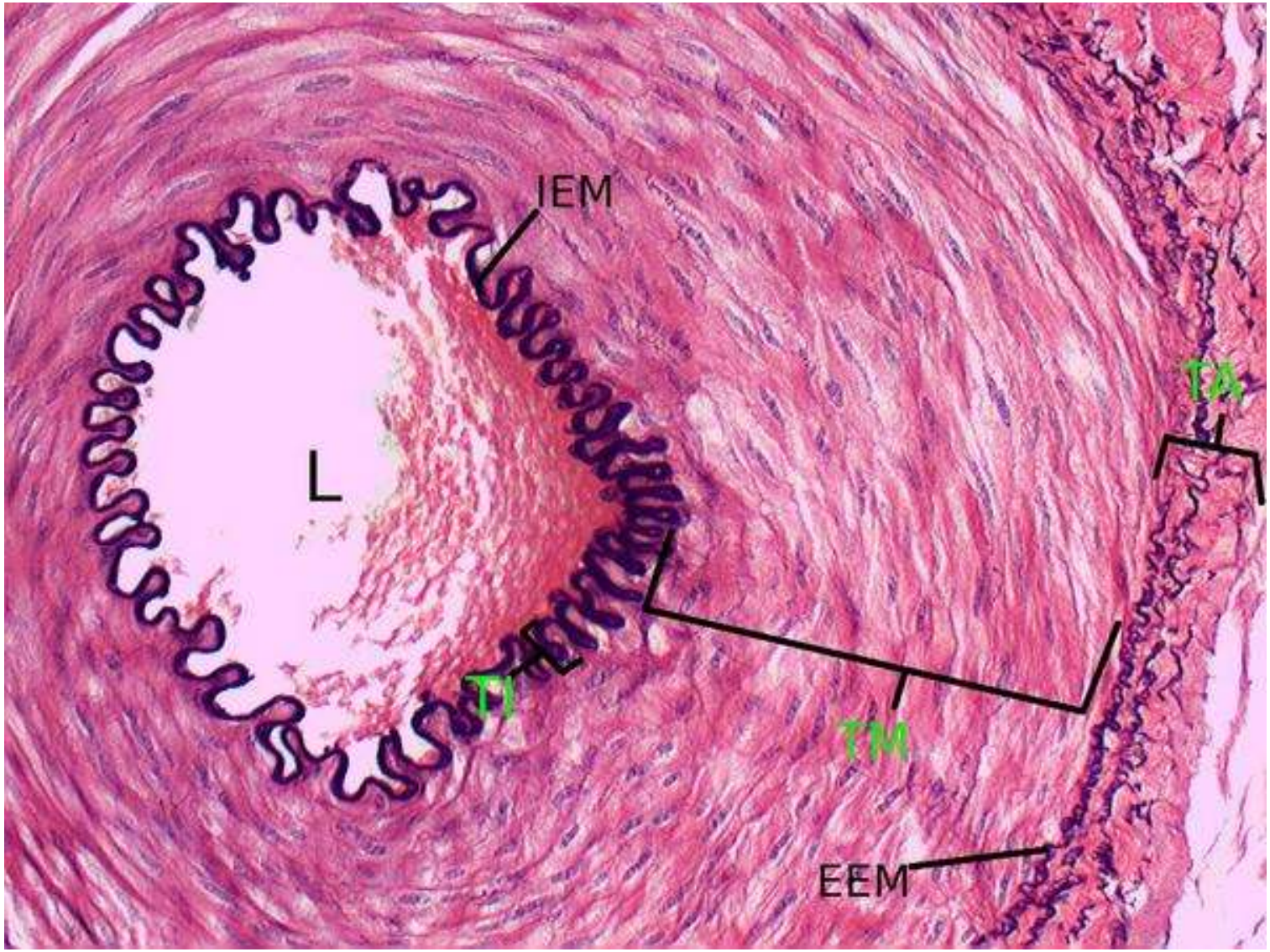
Muscular Artery

T Adventitia:

- ***Elastic and collagen fibres running longitudinally***

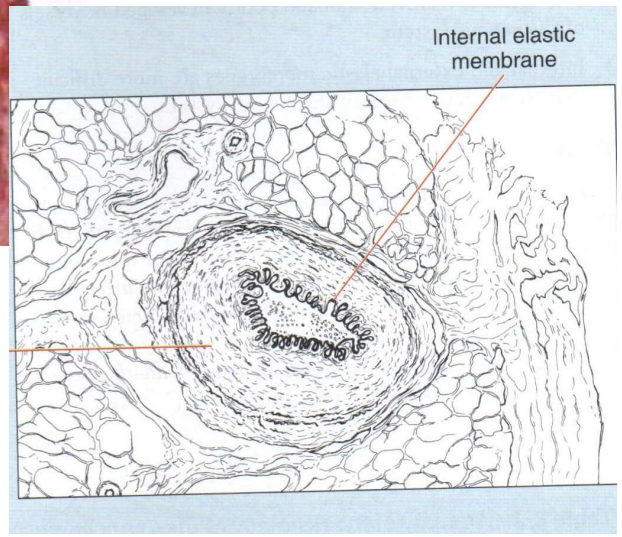


Muscular Artery



Cross section of Muscular artery(100X) stained specifically for elastin





Difference between Elastic & Muscular artery

Elastic Artery

- ***T Intima:*** is much thicker, sometimes 20% of the total wall thickness.
- Subendothelial C T has more elastic fibres
- IEL is not distinct (as it has the same structure as the elastic membranes of the media)
- ***T Media:Elastic fibres ++***

Muscular Artery

- ***T Intim***a: **IEL is well demarcated.**
- ***T Media: Smooth M +***
- **EEL is also very prominent.**
- T adventitia: Thickness is more and sometimes may be equal to the thickness of the media.

Arteriole

- Smallest branches of the arterial system.
- Deliver blood to the smallest blood vessels, the capillaries.
- Diameter - less than 100 Micron m
- Larger Arterioles: Dia 100-50 Micron m
- Terminal Arterioles: Dia less than 50 Micron m
- *Wall is relatively thicker than the lumen.*
- Serve as a transition vessel between muscular arteries and capillaries.

Three coats are present

T Intima: Endothelium +
no subendothelial layer
thin or absent IEL.

T Media: 1-5 layers of circularly arranged smooth muscle cells.

T Adventitia: thin and poorly developed. Made up of a thin network of collagen fibers.

Types of Capillaries

Continuous or Somatic

- **Commonest type**
- **Present in connective tissue, muscle, brain, lung etc**
- **Endothelial cells form the continuous lining of the capillary.**
- **Tight junctions, desmosomes and gap junctions are seen.**

- ***Fenestrated or visceral***
- **Tiny pores(openings , fenestrations) are present in the cytoplasm of endothelial cells.**
- **Pores are often closed by a thin diaphragm(thinner than the cell membranes)**
- **Pores are so designed for *rapid exchange of molecules between blood and tissue(greater permeability)***
- **Examples: Kidney glomeruli intestinal villi, choroid plexus, endocrine glands etc**

§ 3 Types of Capillaries

3. Sinusoids (discontinuous) capillaries-

- Structure– endothelial cells separated by wide gaps; **no basal lamina**
- Conform to the shape of the surrounding tissue
- Molecules can pass– proteins and blood cells
- Locations-- liver, bone marrow, spleen, lymphatic organs
- Fig. 20.7

Venous System

- Veins are thin walled
- Carry blood from capillaries to heart
- Large veins are formed by the union of smaller vein like tributary of a river.
- Often provided with valves
- Valves serve to prevent reflux of blood.

Types of Veins

- Large sized Veins
- Medium sized veins
- Venules
- ***A clear distinction between T intima, media and adventitia cannot be made out in small veins as all the layers consist predominantly of fibrous tissue***

- Same 3 layered organization as arterial system.

- **T intima**
- **T media**
- **T adventitia**

Differences between Medium sized Vein and a medium sized artery

Medium sized vein

- A collapsed **larger lumen**
- **Thin wall**
- **No IEL**
- **T media has large quantity of collagen (**few smooth muscles and less elastic fibers) **that is the reason veins are easily compressed**.
- **T adventitia is thicker than T media in large veins**
- **Presence of valves to prevent back flow**

Medium sized artery

- **Thick wall**
- **Arteries retain their patency**
- **Internal Elastic Lamina is present only in arteries**
- **T media is thicker than adventitia**

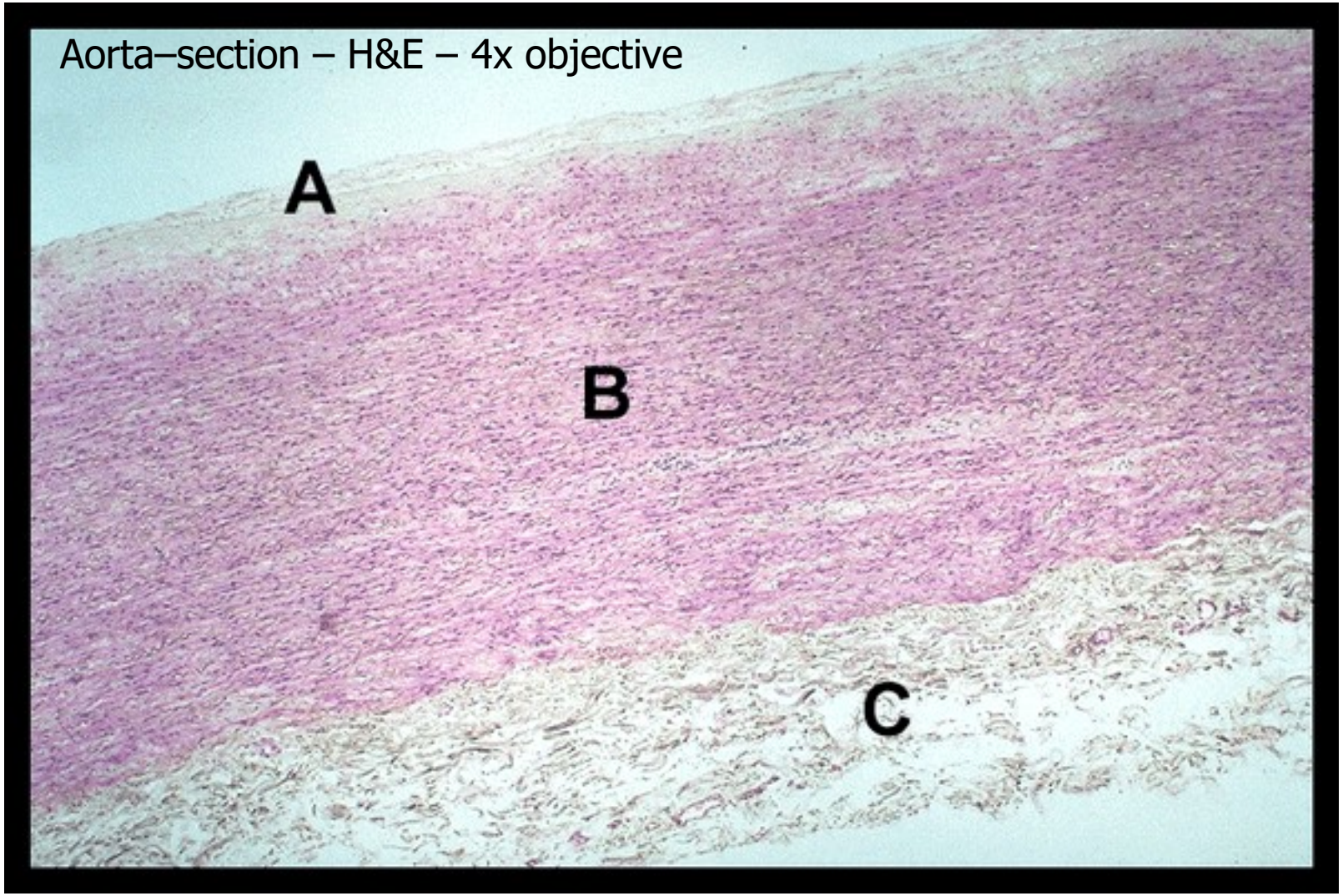
LARGE VEINS

- This group includes the superior & inferior venae cava & the portal vein.
- TUNICA INTIMA:
- The intima of large vein consist of ENDOTHELIUM & a thin layer of SUB ENDOTHELIAL Connective tissue.

TUNICA ADVENTITIA:

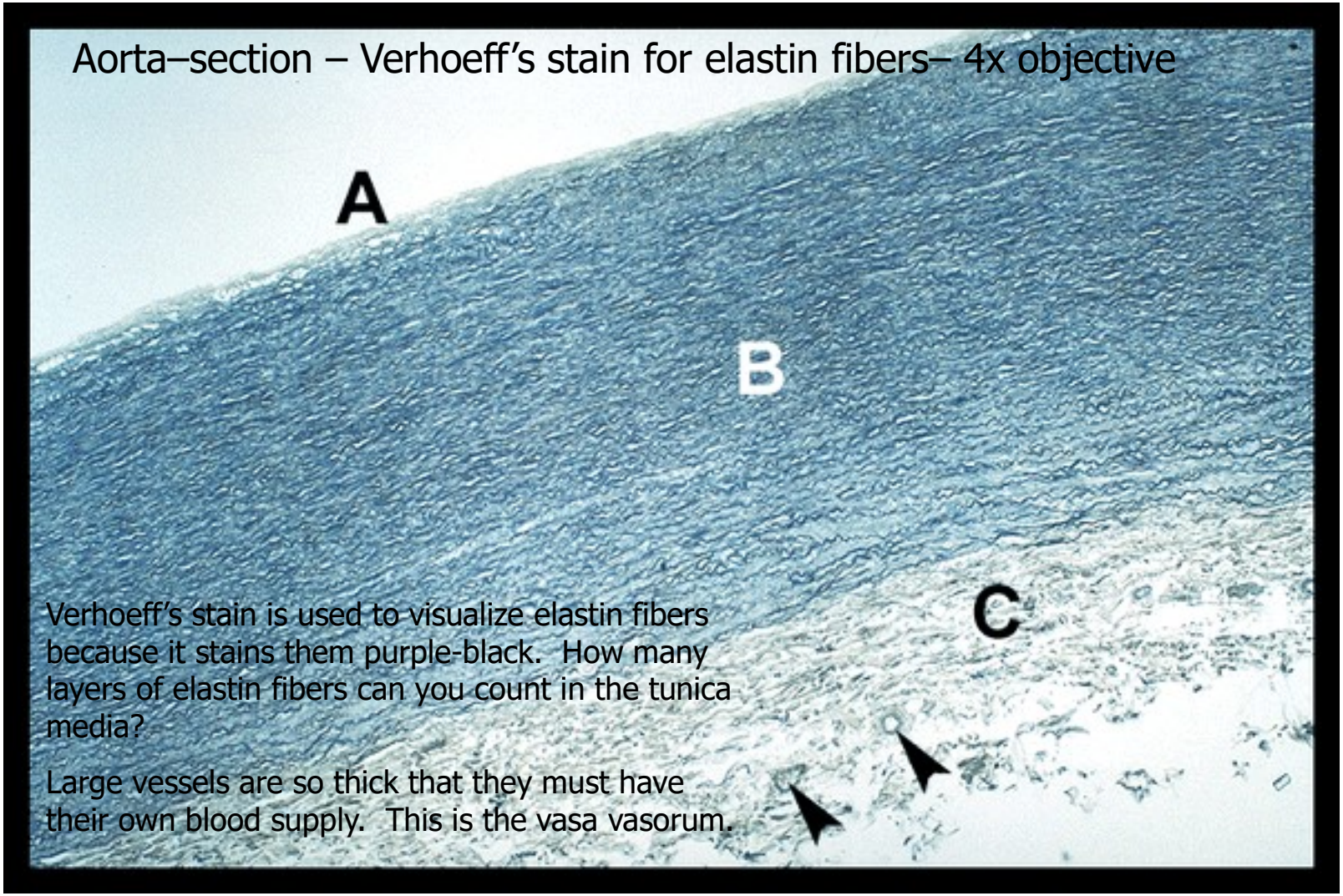
- It is the thickest layer of the three coats & shows three zones.
- Immediately next to the tunica media is a zone containing Dense fibro-elastic connective tissue with course of collagenous fibers running spirally.
- MIDDLE ZONE is composed of longitudinally running smooth muscles cells.
- OUTER MOST ZONE consist of a network of course collegenous & elastic fibers.

Large or elastic arteries



A = tunica intima (interna) B = tunica media C = tunica adventitia (externa)

Large or elastic arteries



A = tunica intima (interna) B = tunica media C = tunica adventitia (externa)
arrowheads = vasa vasorum